Coding & Solution

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/burlaprudhviraj123/OptiCrop-Smart-Agricultural-Production-Optimization-Engine-Smartbridge
Programming Language(s)	Python 3.12, JavaScript (Chart.js), HTML5, CSS3
Framework(s) Used	Flask, Bootstrap 5
Key Features Implemented	• ML Crop Recommendation Model (99.32% accuracy) • Compatibility assessment engine with color-coded progress bars • Multi-crop requirement dashboard (macronutrients, weather, pH)
Pending / Incomplete Features	None (All features fully implemented and verified)
Setup / Run Instructions	1. Run 'venv/bin/python train_model.py' to train classifier model. 2. Run 'venv/bin/python app.py' to launch local web server. 3. Navigate to http://127.0.0.1:5001.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: